Via enclose

Enclosure is a region pattern across layers: inner VIA must sit inside outer metal with margin

The idea

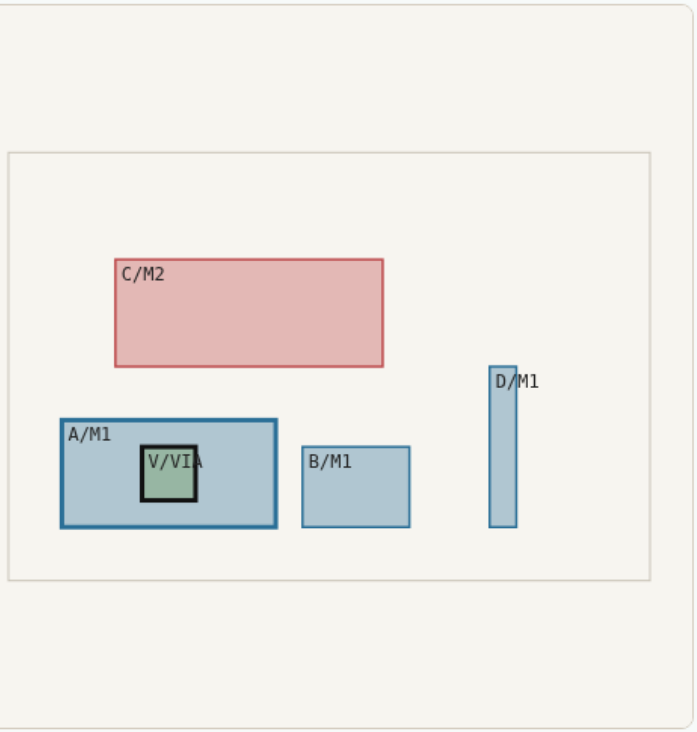
- V inside A on M1 with margins ≥1 → OK
- V not inside C on M2 → enclosure marker
- GOLDEN enclosure_markers=1

Via V

MULTI-LAYER VIA ENCLOSURE

Via V

Step 1 / 5 · v · module03-01-multi-layer-via



Explanation

V is 2×2 on VIA inside A.

- via

2×2

Via V

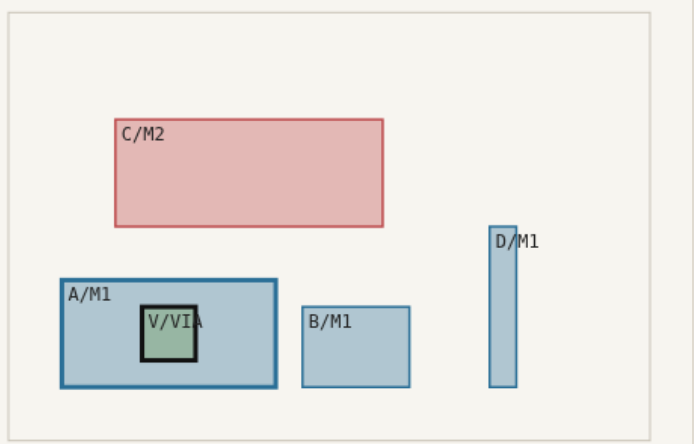
M1 OK

MULTI-LAYER VIA ENCLOSURE

M1 OK

Step 2 / 5 · m1 · module03-01-multi-layer-via

m1_ok



Explanation

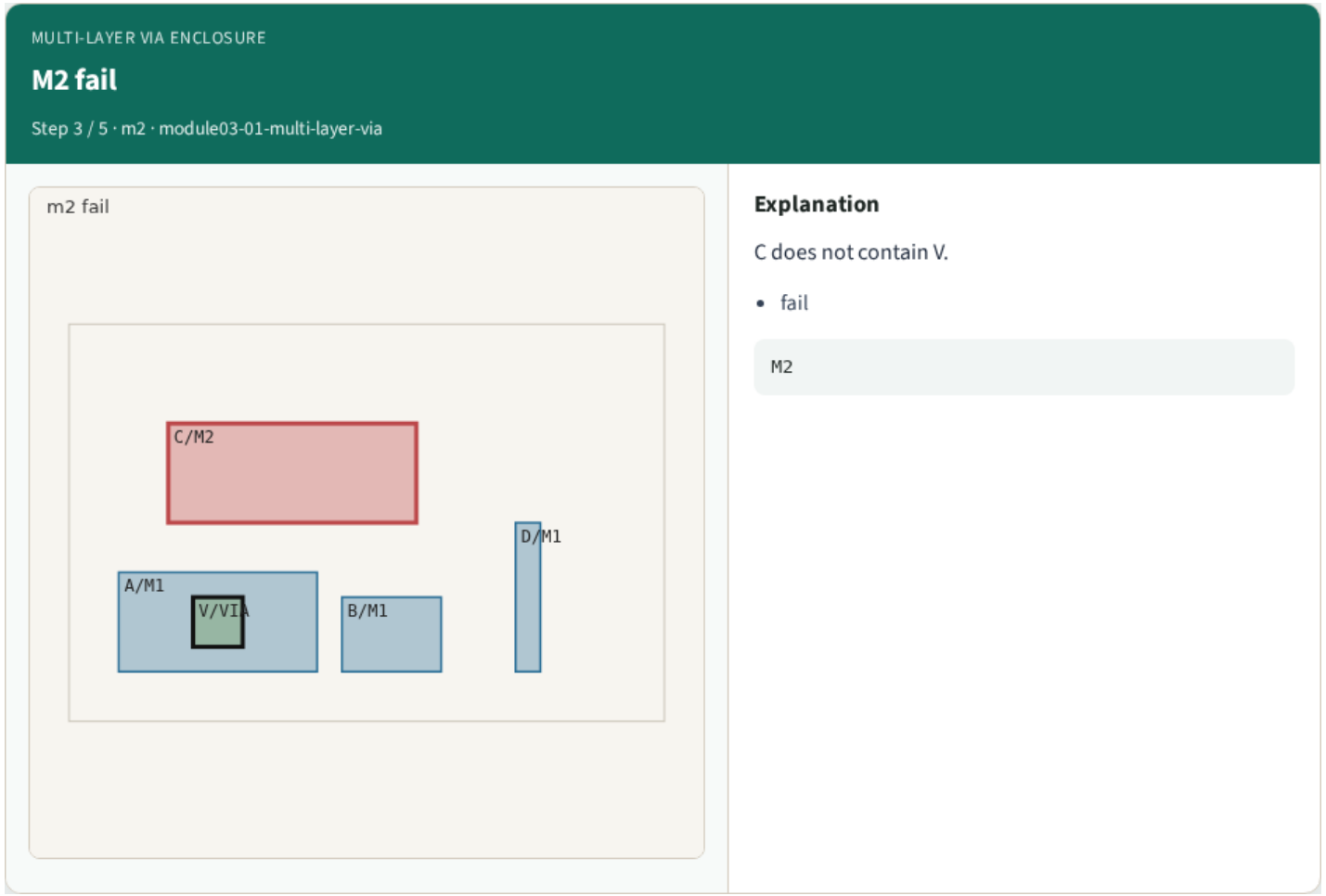
Enclosed by A with margin ≥1.

- encl

```
m1_ok
```

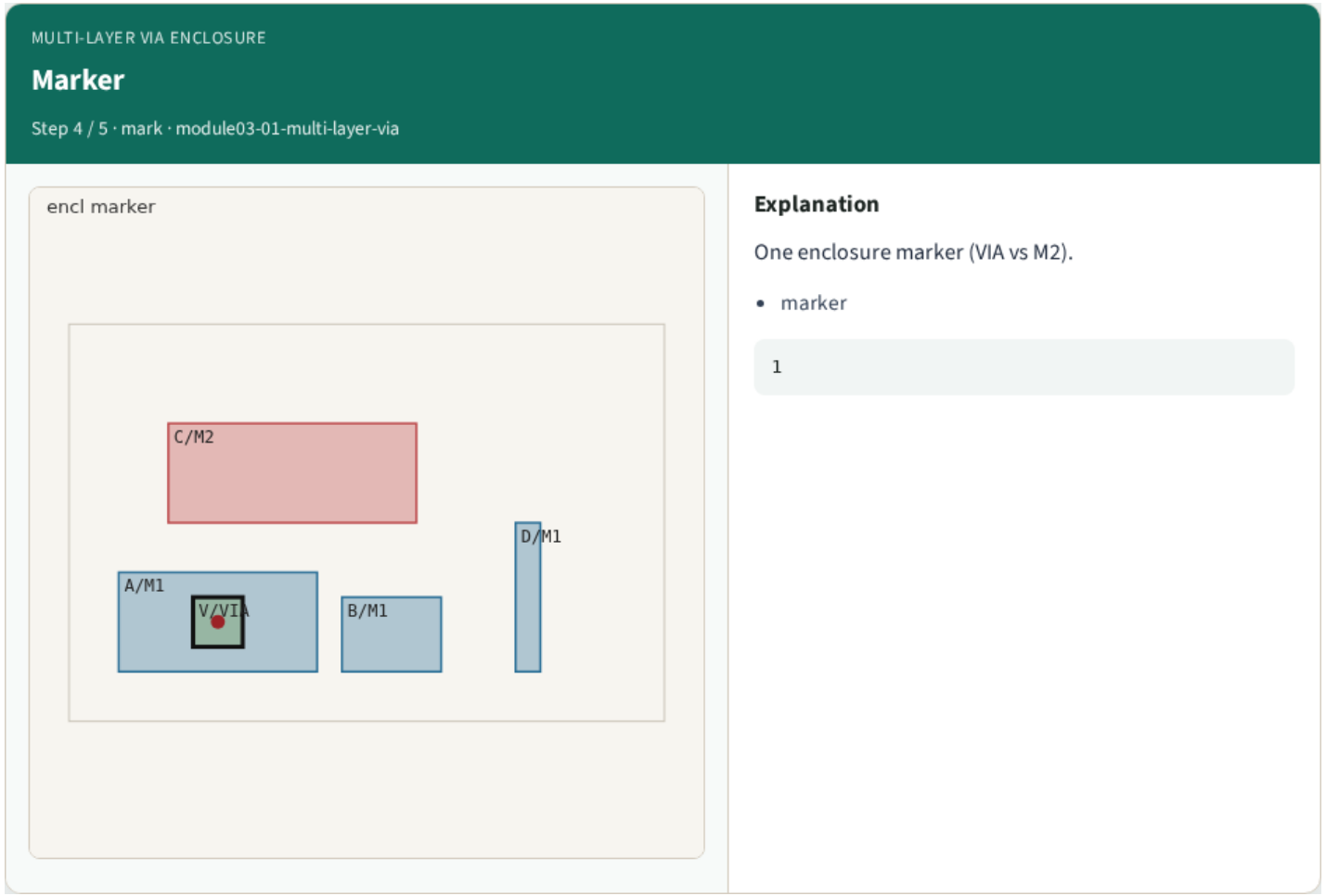
M1 OK

M2 fail



M2 fail

Marker



Marker

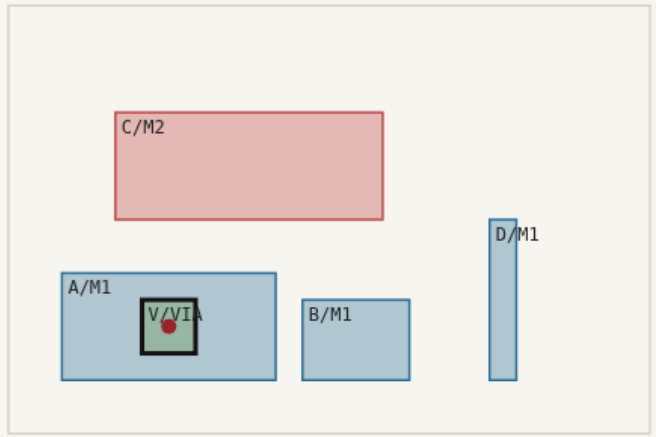
API

MULTI-LAYER VIA ENCLOSURE

API

Step 5 / 5 · api · module03-01-multi-layer-via

m1=true m2=false



Explanation

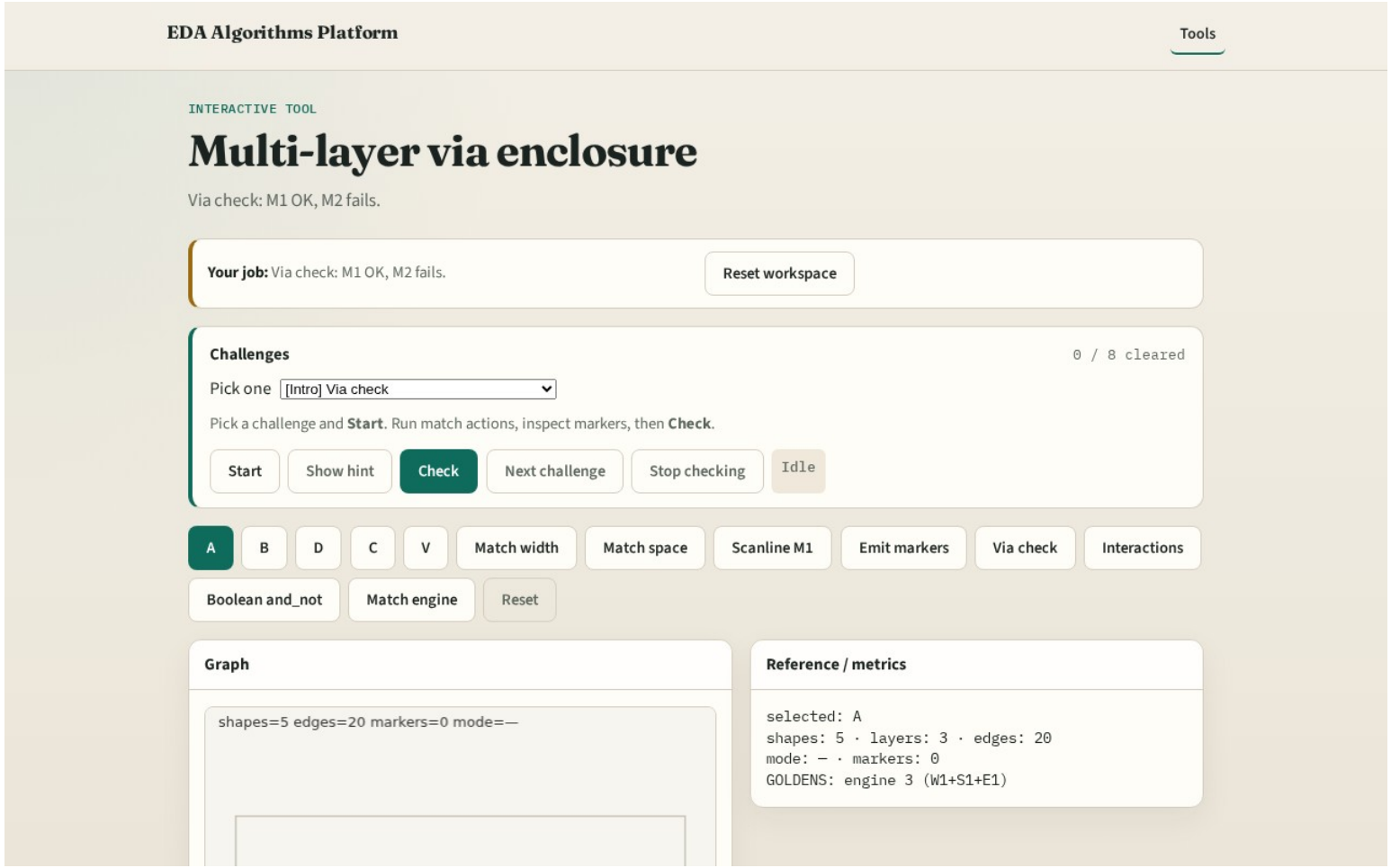
multi_layer_via_check reports m1_ok / m2_ok.

- API

```
E=1
```

API

Browser lab track



Browser lab starter

Implement track

- Call multi_layer_via_check(); assert m1_ok and enclosure_markers==1

Pitfalls

- Requiring same-layer enclosure
- Ignoring margin limit

Your turn

- Next: multi-layer interactions